

KUGAARUK, NU, Canada

WMO: 714070

Lat: 68.538N

Lon: 89.789W

Elev: 16

StdP: 101.14

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-42.7	-41.1	-46.8	0.0	-42.2	-45.4	0.0	-40.9	14.7	-27.5	13.2	-29.0	1.7	60	1.062

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.1	19.4	13.0	17.2	11.9	14.9	10.8	13.5	18.2	12.1	16.4	10.9	14.9	4.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.0	8.2	15.5	9.5	7.4	14.3	8.2	6.7	12.7	38.0	18.3	34.5	16.3	31.4	15.0	18.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.8	11.3	10.1	DB	-45.4	22.9	2.3	2.2	-47.1	24.4	-48.4	25.7	-49.7	26.9	-51.4	28.5	(4)
(5)			WB	-44.9	16.1	1.7	1.5	-46.2	17.2	-47.2	18.1	-48.2	19.0	-49.4	20.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-12.4	-30.7	-31.8	-28.5	-19.1	-7.8	3.3	9.5	7.6	0.8	-7.6	-19.7	-26.8
	DBStd	15.72	5.86	6.17	6.19	6.17	5.29	3.94	3.42	3.45	3.22	4.94	6.85	6.85
	HDD10.0	8253	1262	1171	1193	874	551	206	51	92	275	545	891	1142
	HDD18.3	11236	1521	1404	1452	1124	809	451	273	332	525	803	1141	1401
	CDD10.0	59	0	0	0	0	0	5	36	18	0	0	0	0
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH23.3	3	0	0	0	0	0	0	3	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	4.9	5.2	5.1	5.2	5.0	4.7	4.6	4.4	4.5	4.8	5.1	5.3	5.2
Precipitation	PrecAvg	227	7	6	12	15	13	22	37	42	28	21	10	9
	PrecMax	403	18	12	62	84	59	84	130	123	61	41	22	23
	PrecMin	121	0	0	0	0	0	2	1	6	7	8	2	0
	PrecStd	64	5	3	15	19	12	16	27	25	13	7	4	6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-11.4	-12.5	-9.5	-1.2	4.5	20.1	23.0	21.1	12.4	2.5	-2.4	-5.0
		MCWB	-11.9	-12.9	-10.2	-1.8	2.1	12.9	15.1	14.1	7.8	1.5	-2.9	-5.5
	2%	DB	-15.0	-15.2	-12.6	-4.8	1.9	15.9	19.9	18.4	9.6	1.1	-4.5	-9.6
		MCWB	-15.3	-15.6	-13.1	-5.6	0.7	10.9	13.2	12.9	6.7	0.2	-5.1	-10.0
	5%	DB	-18.7	-18.4	-15.5	-6.7	0.7	12.1	18.2	16.2	7.4	-0.1	-6.9	-13.3
		MCWB	-19.1	-18.8	-16.0	-7.4	-0.2	8.2	12.4	11.8	5.5	-0.9	-7.5	-13.7
	10%	DB	-21.7	-22.5	-18.2	-8.9	-0.6	9.2	16.1	13.7	5.6	-1.3	-9.5	-16.6
		MCWB	-22.0	-22.7	-18.6	-9.6	-1.6	6.4	11.1	10.3	4.1	-1.9	-9.9	-16.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-11.7	-12.9	-10.1	-1.9	2.6	12.9	16.2	15.0	8.4	1.7	-2.8	-5.4
		MCDB	-11.4	-12.7	-9.6	-1.1	4.3	19.6	21.2	19.8	11.1	2.4	-2.4	-5.0
	2%	WB	-15.4	-15.7	-13.1	-5.5	0.8	10.8	14.1	13.3	6.9	0.3	-5.1	-10.0
		MCDB	-15.2	-15.3	-12.6	-4.9	1.7	15.6	19.0	17.5	9.3	0.9	-4.6	-9.6
	5%	WB	-19.1	-18.7	-16.0	-7.4	-0.1	8.4	12.6	12.0	5.6	-0.8	-7.4	-13.7
		MCDB	-18.8	-18.3	-15.6	-6.8	0.8	11.8	17.1	15.5	7.3	-0.2	-6.9	-13.4
	10%	WB	-21.9	-22.7	-18.6	-9.5	-1.5	6.5	11.3	10.6	4.2	-1.9	-9.9	-16.9
		MCDB	-21.7	-22.5	-18.2	-8.9	-0.6	9.1	15.7	13.7	5.6	-1.4	-9.5	-16.6
Mean Daily Temperature Range	5% DB	MDBR	6.5	6.5	7.6	8.5	6.0	5.8	8.1	7.1	4.2	4.9	6.3	6.3
		MCDBR	9.4	9.5	10.7	8.6	5.3	11.1	11.8	11.4	8.1	5.0	7.8	8.5
		MCWBR	9.3	9.3	10.4	8.3	4.7	7.2	6.6	6.8	5.9	4.9	7.7	8.4
	5% WB	MCDBR	9.3	9.4	10.7	8.5	5.3	10.8	10.9	10.5	7.8	4.9	7.5	8.3
		MCWBR	9.2	9.2	10.5	8.2	4.7	7.1	6.7	6.7	5.9	4.9	7.4	8.2
Clear-Sky Solar Irradiance	taub	0.149	0.206	0.255	0.301	0.327	0.323	0.357	0.325	0.295	0.239	0.156	N/A	
	taud	1.886	2.074	2.167	2.095	2.104	2.337	2.377	2.489	2.450	2.088	1.904	N/A	
	Ebn at Noon	113	607	784	835	852	875	827	822	750	518	84	0	
	Edh at Noon	6	44	77	113	127	104	98	78	61	42	5	0	
All-Sky Solar Radiation	RadAvg	0.03	0.44	2.00	4.41	6.14	6.04	5.16	3.48	1.69	0.53	0.08	0.00	
	RadStd	0.00	0.03	0.07	0.15	0.35	0.51	0.50	0.27	0.14	0.06	0.01	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	+0.91	+0.95	N/A	N/A	N/A	N/A	-436	-432	N/A	N/A

Nomenclature: See separate page